

MODULE-1
OVER VIEW OF CIVIL ENGINEERING

SCOPES OF CIVIL ENGINEERING:

1. Scopes of Structural Engineering: Following are the scopes Structural Engineering

- Analysis and design of Dams, Bridges, Stadiums, Auditoriums, Multi - storied Buildings.
- Analysis and design of power generation stations
- Analysis and design of steel industrial structures
- Repair, rehabilitation and maintenance of structures
- Design of nuclear Power plants.
- Design of structural reinforcement for different type of structural components

2. Scopes of Geotechnical Engineering:

- To assess the quality and strength of soil or rock to construct civil engineering structure
- To design retaining walls for soil retainment.
- To decide type of foundations for different type of structures.
- For design of underground structures such as Tunnels, Shafts, Conduits.
- Design of earthen dams for storage of water.
- For the design of Roads for transportation facilities

3. Scopes of Construction Technology:

- Planning for Construction Equipment and Machinery for construction project.
- Deals with planning, scheduling and execution of construction activity related to a project.
- Comprises of men, material, time and money management.

- Emphasis will be on new construction practice such as pre-cast construction and pre stress constructions, safety of men and material, utilization of marginal materials etc.
- Construction of underground Tunnels using modern equipment's.
- Construction of floating structures.

4. Scopes of Hydraulics

- To measure the discharge of water in rivers for design of bridges
- Design of hydro power plants for generation of electricity.
- Design of Pumps and turbines
- Design of water supply schemes for the city which includes design of pipes and pumps.
- Design of canals to carry water to irrigation land from dams.
- Design of Weirs for Dams

5. Scopes of Water Resource & Irrigation Engineering **Scopes of Water Resource Engineering**

- To arrive the total discharge of water from rainfall from catchment areas
- To design the reservoir capacity to store the water
- Water quality management and pollution control.
- Scope for usage of water for garden and recreational centers.
- Design of water supply systems for the cities and industries.
- Flood mitigation, land drainage and culverts for control of water.

6. Scopes of Irrigation Engineering

- To understand system of irrigation and command area development aspects
- To supply the water to the irrigation land to grow the crops as per the requirement
- To improve the ground water storage of the land.

- To increase the food production by supplying optimum quantity of water to the crops
- Additional revenue to the nation in the form of water tax and increased yield in crops.
- Cultivation of commercial crops such as Sugarcane, Cotton etc

7. Scope of Transportation Engineering

- It involves planning, design, construction, operation and maintenance of transportation facility.
- Planning and design of air strip runways, roads, harbors and railways
- Maintenance and up gradation of harbors, airports, railway system based on requirements.
- It contributes economic, industrial, social and cultural development of any country.
- Design of traffic signals for control of traffic.

8. Scope of Environmental and Sanitary Engineering

- Involves collection of water, Purification and supply for drinking.
- Waste water collection, treatment and disposal
- Air pollution control and treatments.
- Solid waste management and control
- E-Waste management control and Treatment
- Construction waste management and control

9. Scope of Geographic information System (GIS)

GIS is a computer-based information system that performs the activities of capturing, storing, manipulating, analyzing and displaying both spatial and non-spatial data in order to solve complex research, planning and management problems. Following are the uses of GIS

- Telecom and network services
- Accident analysis and hot spot analysis
- Urban Planning
- Transportation Planning
- Determine land use/land cover changes
- Navigation (routing and scheduling)
- Flood damage estimation
- Natural Resources Management

10. Scope of Earthquake Engineering

- An earthquake is a sudden slipping or movement of a portion of the Earth's crust or plates, caused by a sudden release of stresses
- The study of the behavior of buildings and structures subject to seismic loading
- To forecast the potential consequences of strong earthquakes on urban areas and Civil infrastructure
- Design, construct and maintain structures to perform at earthquake exposure.
- It also assists analyzing the interaction between civil infrastructure and the ground, including the consequences of earthquakes on structures.
- Design of seismic control devices to minimize the seismic effects on buildings and other infrastructure

Building Materials

i) Stone :

A piece of rock that is used in construction is called stone. Following are the Uses of stones:

- Stones are used for the base preparation of different constructions and also they are important for general building works like in foundation, floors covering, masonry works etc.
- Stones are used for the ornamental and architectural beauty of different types of structures
- Generally, they are used for railway ballast and as an aggregate in cement and lime concretes.
- Stone are used for making fireproof structures
- Stones are used as a construction material of different structures likes dams, columns, bridges piers, harbors etc.
- Limestone acts as a flux in the manufacturing cement and lime

ii) Brick:

Brick is a small rectangular block typically made of fired or sun-dried clay, used in construction. Following are the Uses of Bricks

- The first class and second class bricks are used for permanent structures such as buildings, dams, roads, bridge piers, etc
- The first class bricks are also used for architectural effects on structures
- The second class bricks are used in construction but the surface should be plastered evenly because of the irregular size and shape of bricks
- The third class or sun-dried bricks are used only for temporary structures
- Broken bricks are used as aggregate for foundation concrete and road metal works.

iii) Wood:

Wood is the material which forms the trunks and branches of trees. Following are the uses of wood

- In construction of houses, wood is used commonly used for flooring, frames of doors and windows
- For artworks such as statues, carvings and making decorative objects wood is used
- wood is the best for making furniture
- Hardwood and softwood were used in the past for ship industry
- Wood is an age-old source of energy all over the world

iv) Glass:

Glass is a hard, brittle substance, typically transparent, made by fusing sand with soda and lime and cooling rapidly. Following are the uses and properties of glass.

- It is most typically used as transparent glazing material in the buildings, including windows in the external walls
- Glass has been used to enhance the aesthetic view of structure
- Glass Wall Cladding provides a sleek and easily maintained wall surface and also increases aesthetic view of structure
- A glass is 100 percent recyclable i.e. it does not degrade during the recycling process.
- Glass can withstand the effects of water, wind, sun and other environmental effects, without losing an inch of its appearance and integrity.

v) Aluminum:

A chemical element that is a light, silver- colored metal, used especially for construction and aircraft parts. Following are the uses of Aluminum

- Aluminum sections are used for the construction of high rise buildings due to its high strength to weight ratio
- Aluminum frames are generally durable, cost-effective hence they are used for partitions, window frames in buildings.
- Aluminum is useful in aviation industry because of three excellent properties such as high strength to weight ratio, excellent ductility, and high resistance to corrosion
- The advanced aluminum alloys are used in spacecraft and rocket technology.
- The aluminum is used in the ship constructions.

vi) Cement:

Cement is a powder of alumina, silica, lime, iron oxide, and magnesium oxide burned together in a kiln and finely pulverized and used as an ingredient of mortar and concrete. Following are the uses of cement.

- The most significant use of cement is production of concrete and mortar
- Cement mortar can be used for masonry work, plaster, pointing, etc.
- It is used in concrete for laying floors, roofs and constructing lintels, beams, stairs, pillars etc.
- It can be used for preparation of foundations, watertight floors, footpaths, etc.
- It can be used for manufacturing precast pipes, garden seats, flower pots, dust bins, fencing posts, etc.

vii) Aggregates:

Aggregate is a granular material, such as sand, gravel, and crushed stone.

Following are the uses of aggregates.

- Increases the volume of concrete, thus reduces the cost.
- Aggregates are used to provide a rigid structure
- To reduce the shrinkage and cracking in concrete
- Aggregates are used in many structures and substructures e.g. different elements of a Building, bridges, foundations etc.,

Vii) Concrete:

Concrete is a construction material composed of cement, fine aggregates (sand) and coarse aggregates mixed with water which hardens with time. Following are the Uses of Concrete.

- Concrete is used in the construction of residential Buildings
- Concrete is used in the construction of roads and Expressways.
- Concrete is used in the construction of Marine constructions
- Concrete is used in the construction of Culverts and sewers
- Concrete is used in the construction of Water tanks
- Concrete is used in the construction of Foundations

viii) Steel:

Steel is a kind of metal alloy that's made of iron and carbon. Due to its high strength it is often used to build the framework of skyscrapers. Following are the uses of steel

- Steel is used as reinforcement in reinforced concrete structures.
- Steel is used to construct high rise buildings.
- Steel is used to construct industrial buildings
- Steel is used to construct Residential buildings
- Steel is used to construct Bridges

ix) RCC (Reinforced Cement Concrete):

Reinforced Cement Concrete (RCC) is a composite building material consisting of concrete reinforced with steel. Following are the uses of Concrete

- Reinforced concrete is used for the construction of roof slabs, columns, beams and footings in residential and commercial structures.
- Reinforced concrete is used for construction of bridges of small, medium and long spans resulting in aesthetically superior and economical structures in comparison with steel bridges
- Reinforced concrete is used in the construction of roads that is designed to carry heavy traffics loads.
- Reinforced concrete is used in the construction of almost all types of foundations of structures.
- Pipes and conduits have been constructed from reinforced concrete

x) PSC (Pre-stressed concrete):

Pre-stressed concrete is a structural material that allows for predetermined, engineering stresses to be placed in members to counteract the stresses that occur when they are subject to loading. Following are the uses of PSC

- Pre-stressed concrete is used in the construction of superstructures of bridges
- Pre-stressed concrete is used in the construction of runways
- It is used in the Construction of Water tanks.
- Pre-stressed concrete is used in the construction of tall columns and retaining walls

xi) Smart Materials:

Smart materials have different properties that can be changed according to the changes in the external factors, such as temperature, light, stress, moisture etc.,.Following are the benefits of Smart Materials.

- Smart concrete technology offers an alternative method for monitoring the health of reinforced concrete structures.
- Crystalline water proofing of concrete to increase the efficiency of curing.
- Increases Concrete durability
- To reduce corrosion of steel

Role of Civil Engineer in the Development of Nation:

- Civil engineer has very important role in the development of the infrastructures like new townships, building structures for the rural and urban areas for various utilities.
- Construction of reservoir's, Canals and distributaries to take water to agricultural fields
- Provide and maintain communication systems like roads, railways, harbors and airports.
- Purify and supply water to the areas like houses, schools, offices etc.
- Monitor land, water and air pollution and take measures to control them
- Improved irrigation facility enhances agricultural products and hence producers as well as consumers are benefitted.
- He is responsible for solid waste management
- A well planned and built network of roads and road crossings.
- Infrastructural facility develops scope for a number of industries and it creates job opportunities
- Construction of Primary health centers, specialized hospitals add to the desirable infrastructure facility.

Question Bank (For Internal Test)

- 1) List the scopes of following Civil Engineering Fields.
 - a) Structural Engineering
 - b) Geotechnical Engineering
 - c) Construction technology
- 2) Explain scopes of following Civil Engineering Fields
 - a) Hydraulics
 - b) Water resource and Irrigation engineering
 - c) Transportation engineering
- 3) Explain scopes of following Civil Engineering Fields
 - a) Environmental and Sanitary Engineering
 - b) GIS
 - c) Earthquake engineering
- 4) Explain the uses of following construction materials
 - a) Stone
 - b) Bricks
 - c) Wood
- 5) Write short note on
 - a) Glass
 - b) Aluminum
 - c) Cement
- 6) Explain the uses of following building materials
 - a) Aggregate
 - b) Concrete
 - c) Steel
- 7) Explain the uses of following building materials
 - a) RCC
 - b) PSC
 - c) Smart materials
- 8) Explain the role of Civil engineer in the development of nation.